What Reaches the Graph Decides Held-Out Question Answering in Knowledge Graph Construction

Minwoo Kim, Philje Bang†, Junpyo Lee†, Mingyu Jeon†, Hyun Kyeong Lee†, Hail Jung, and Hongdon Lee

Abstract—Language models build knowledge graphs from text, but it is unclear which parts of a competency-question (CQ) pipeline make a graph answer unseen questions. We varied the pipeline one component at a time and scored the graphs on held-out questions, on two corpora and two model families, with CQs generated from the corpus text. A relation vocabulary induced from the corpus, without gold knowledge, raised single-pass answer correctness by 16.3 points on the multi-hop corpus and 12.5 on the second, and a vocabulary induced outside the evaluation documents also raised it. With a 22-name relation list, extraction from every chunk without CQs reached the combined pipeline's grounded question answering (QA) at 106k tokens per unit, against 277k for the combined pipeline's first pass and 838k for its full run, and with more precise facts. Without any vocabulary, the same extraction scored .558 grounded QA against .430 for iText2KG, .445 for EDC and .213 for AutoSchemaKG, three public extractors. CQs kept their value in two places. On fresh CQs run as structured queries, the combined pipeline passed 13.5 points more than that extraction. With the list and at equal fact count, the CQ-guided pass scored 14.3 to 25.0 points above model-scored and chain-preserving selections of the listed extraction. A corpus-induced relation vocabulary with extraction from every chunk is a simple, strong baseline for CQ-driven graph construction. We release plans, code and graphs.

Index Terms—Competency questions, knowledge graph construction, question answering, relation vocabulary.

1 Introduction

Published competency-question (CQ) driven knowledge-graph (KG) pipelines change several components at once and report one final score, so it is unknown which component improves the finished graph's answers to questions it never saw during construction. We varied the components one at a time and found that held-out question answering (QA) follows what reaches the graph. Grounded QA is the share of questions answered correctly with support from retrieved graph facts. Three questions organise the study. RQ1: does a relation vocabulary, a list of 22 relation names given to the extractor, raise held-out QA, and does it need gold knowledge? RQ2: given that vocabulary, does extraction without CQs reach the grounded QA of the combined pipeline and of public extractors? RQ3: at the same fact count, does CQ guidance beat selections made without CQs? The answers make a corpus-induced relation vocabulary with extraction from every chunk a simple, strong baseline for CQ-driven graph construction. With a relation list, that extraction reached the combined pipeline's grounded QA at 106k tokens per unit, against 277k for the combined pipeline's first pass and 838k for its full run.

Nothing guarantees that a graph a large language model (LLM) pipeline builds from text can answer the questions people will ask. Ontology engineering approaches this with competency questions [1], [2]. A CQ states something the ontology must answer, so a set of CQs scopes and tests it. In this pipeline a CQ is a question generated from a chunk, with an expected answer span. Recent pipelines generate CQs from text, grow a schema from them, run them against the graph and repair what fails, a cycle we call the repair loop (Section 2).

Consider the question "What is the cause of death of director of film Wedding Bills?" from 2WikiMultihopQA [3]. According to the source text, the director, Erle C. Kenton, died in Glendale "from Parkinson's disease". With Qwen, the graph built by a single-pass pipeline with no relation list holds the edge from the film to Kenton but no cause-of-death triple for him. The answerer says "unknown". Given a list of 22 relation names that includes "cause of death", the same pipeline extracts the triple and answers correctly. CQ count, repair rounds and token budget could do the same, so we varied each alone.

Published CQ-driven pipelines [4], [5], [6] and repair loops [7], [8], [9] vary several of these components together, so an arm that reads more text or writes more facts can win without any component helping. Prior schema-guided and schema-inducing extraction shows that a schema helps extraction [10], [11], but they vary the schema, not the relation vocabulary of a CQ pipeline at equal volume.

We ran seven experiments and one error analysis (Ta-

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ble 2) on two corpora and two model families, Qwen and Gemma. Most contrasts use Qwen on 2WikiMultihopQA, and Gemma repeated some. Four pipelines recur: the single pass (L0), the loop (LOOP), extraction without CQs or a schema (U-ext) and our combined pipeline, all defined in Table 1 (Section 3.1). Answer correctness drops the support requirement of grounded QA, and code scores both against gold (Section 3.2). Held-out questions stay out of every construction stage, and arms use the same tokens, patch size or fact count wherever volume could confound the comparison. Plans with hypotheses and decision rules were fixed before the main runs (Section 3.2).

We report one result per question. Table 3 lists each contrast with its status. The relation list raised every pipeline, and guided repair was not above random patches of the same size.

1) *RQ1.* A relation vocabulary lifts the single pass, and an induced one needs no gold knowledge. A list induced from corpus chunks raised answer correctness by 16.3 points on 2WikiMultihopQA and by 12.5 on T-REx. Lists induced only from passages outside the evaluation shards raised the single pass on both corpora, and the effect replicates with Gemma (Section 4).
2) *RQ2.* With the relation list, extraction from every chunk without CQs (seeded extraction) reached the combined pipeline's grounded QA at 106k tokens per unit, against 277k for the combined pipeline's first pass and 838k for its full run, and with more precise facts. Without any list it scored above three public extractors, iText2KG, EDC and AutoSchemaKG, which also ran without a list. Against iText2KG it scored .558 to .430 (Section 5). With the list, the graph also answered more questions correctly than retrieval over the same chunks. On 40 fresh CQs per shard run through SPARQL, the combined pipeline passed 13.5 points more queries than seeded extraction ([10.7, 16.3]): held-out QA is reached by the simple baseline, and the CQ pipeline leads on structured queries asked through CQs.
3) *RQ3.* With the relation list and an equal fact count, CQ-guided facts scored higher than selections made without CQs. The guided single pass scored 14.3 points above a model-scored selection and 25.0 above a chain-preserving selection of seeded extraction on the multi-hop corpus. It also scored 39.1 points above a random subset, and the lead held on T-REx and against heuristic, CQ-based and chunk-balanced selections. Without the list, the relation-frequency selection scored 8.0 points above the guided pass.

On method, we score recall under a per-question triple budget (capped recall, computed as a hypergeometric expectation) and compare arms at equal token totals and equal fact counts. We release plans, deviation logs, code and graphs.

2 Related Work

Competency questions and LLM-assisted ontology engineering: Competency questions fix the scope of an ontology before it is built [1], [2]. LLM tools now elicit CQs (OntoChat [12]), draft OWL ontologies from CQs (Ontogenia [13]) and verify CQs against an ontology (OE-Assist [14]). IDEA2 keeps an expert in the loop [15].

From CQs to knowledge graphs: Feng et al. derive relations from CQs and ground graph construction in the resulting ontology [4]. HyDRA builds an ontology from CQs that agents agree on [5], and SocraticKG expands each passage into question–answer pairs before extraction [6].

Iterative repair and prompt optimisation: TCR-QF, EvoOntology and OaK repair a graph or an ontology over several rounds [7], [8], [9]. A controlled study of ontology generation from insurance contracts found no gain from iterative repair in a single-agent baseline [16]. DSPy, MIPROv2 and GEPA optimise prompts against a metric [17], [18], [19] and have been applied to triple extraction and ontology learning [20], [21].

Text-to-KG extraction and KG-augmented QA: iText2KG builds graphs incrementally without a fixed schema [22], and we compare it with our extraction in the same setup (Section 5). KGGen [23] measures extraction, Text2KGBench scores extraction when the ontology is given [24], and GraphRAG combines an entity graph with community summaries for retrieval [25].

Schema-guided extraction, schema induction and QA-based evaluation: CodeKGC encodes a pre-defined entity and relation schema in code prompts [26]. KnowCoder represents schemas, among them 876 relation types, as code classes that the model follows [27], and GoLLIE reports that label definitions and guidelines in the schema are central to zero-shot extraction [10]. EDC (Extract, Define, Canonicalize) runs with a pre-defined target schema or one it induces and canonicalises itself [28], and we score that self-induced mode in our setup (Section 5). AutoSchemaKG induces a schema from text and reports multi-hop QA gains [11]; we score it in our setup as well (Section 5). HippoRAG scores a schemaless OpenIE graph by multi-hop QA [29]. These studies show that a supplied or induced schema steers extraction [10], [11], [27], [28]. Within one pipeline at equal volume we separate the relation vocabulary from the competency-question machinery. A vocabulary induced from the corpus without gold carries the gain on two corpora. With a relation list, extraction without competency questions is not distinguishable from the combined pipeline's grounded question answering and uses fewer tokens (Section 5).

3 Pipeline and Experimental Design

3.1 The Curation Pipeline and Its Components

All experiments use one pipeline family, which reads a corpus split into chunks and produces a Web Ontology Language (OWL) style schema (TBox) and a set of facts (ABox) stored as Resource Description Framework (RDF) triples. Formally, a constructed knowledge graph is represented as $G = (V, E)$, where V denotes the set of entity vertices and $E \subseteq V \times \mathcal{R} \times V$ denotes directed relational edges labeled by predicates from a relation vocabulary $\mathcal{R}$. Figure 1 shows the five stages, and Table 2 lists the studies, Exp1 to Exp8.

Stages:

1) *CQ generation.* The generator reads a batch of chunks and writes questions that a reader of those chunks

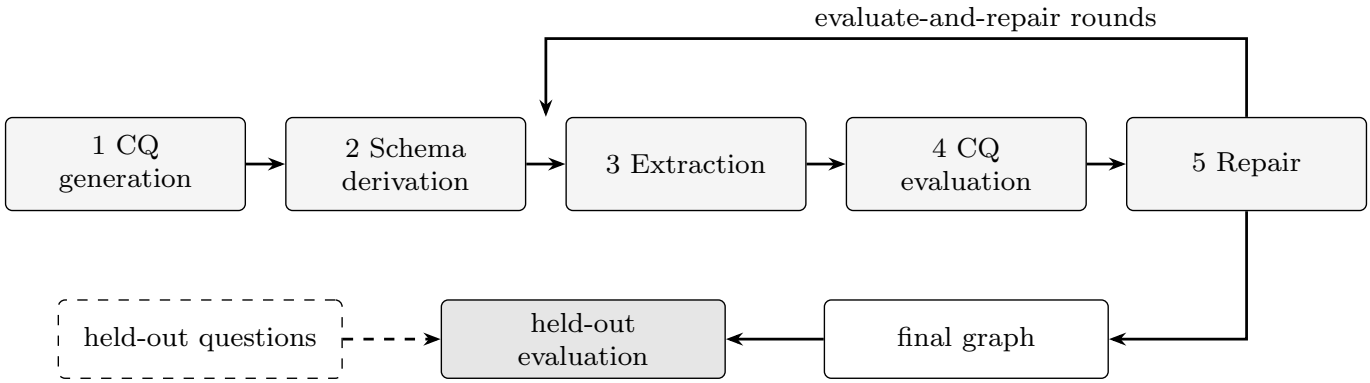


Fig. 1. The pipeline builds the graph in five stages, and evaluate-and-repair rounds loop from repair back to schema derivation and extraction. Held-out questions reach only the evaluation of the final graph.

should be able to answer. Each CQ, a question generated from a chunk, keeps its source chunk and an expected answer span, which code checks against the chunk.

2) *Schema derivation.* Classes and properties are derived from the CQs.
3) *Extraction.* Facts are extracted chunk by chunk, guided by the current schema. Every fact keeps the chunk it came from.
4) *CQ evaluation.* Each CQ is translated to SPARQL Protocol and RDF Query Language (SPARQL) queries and run against the graph. A deterministic rule first checks that the expected span appears in the result rows. An LLM judge then grades the answer 0, 1 or 2.
5) *Repair.* Failed CQs are attributed either to the query or to the graph. Graph-side failures produce patches (added facts, merged entities, corrected values or new properties). A gate scores each patch on a reserved set of curation CQs, never on held-out questions, and accepts or rejects it.

Components under test: Every study except the Exp6 error analysis changes one of the following components or removes one from the combined pipeline.

Relation list. A relation list from the training split: the 22 most frequent relation types (share of at least 0.1%) in the evidence inventory of the 2WikiMultihopQA training split, where no evaluation question was used. The names cover the relations of 98.4% of the 1,216 gold evidence items of the evaluation set. Only the names are given, in alphabetical order, with no frequencies or examples. The names enter the prompts as one block with the same wording and position in every arm with the list (Table 1), and Appendix A.1 lists them.

CQ coverage. Coverage is the share of chunks that are the source of at least one CQ. A first round writes a fixed number of CQs (40 in most experiments). *Expansion* adds CQs over several rounds, each time drawing from chunks that are not yet the source of any CQ, until every chunk is covered.

CQ count. Exp8 varies the number of CQs at a fixed or increased token budget.

Evaluate-and-repair rounds. Stages 4 and 5 run for several rounds. The loop used in Exp5, and as the LOOP baseline in Exp7, repairs queries first and patches only graph-side failures.

Budget and extraction passes. The total number of tokens, and how the remainder is spent. In the combined pipeline, tokens left after expansion go to additional full-corpus extraction passes that ask for facts missing from the current graph.

Where the 22 names come from: The original list is the set of relation types with a share of at least 0.1% in the evidence inventory of the training split, which gives 22 names. An induced list uses no question and no gold relation. The extraction model reads 300 randomly sampled passages (title and first 1,200 characters) in 20 batches of 15 and proposes up to 15 relation names per batch, and the names are normalised to one to four lowercase words. The 80 names proposed in the most batches go to one model call that merges near-duplicates and keeps exactly 22. Induced lists have 22 names so that they equal the size of the original list and differ only in content (Appendix A.1).

Table 1 defines every arm. The *combined pipeline* ("Combined" in tables) starts from the relation list, and a *+vocab* suffix on any other arm name means the same pipeline run with that list.

3.2 Experimental Design

Data: Dataset questions are held out from every stage of construction and only score the final graph. No loop, gate or prompt sees them. The pipeline generates and uses its own curation CQs freely.

The main corpus is 2WikiMultihopQA (2Wiki) [3]. A *shard* is a group of questions together with the union of their context paragraphs, with 8 questions in Exp4 and 16 in Exp5, Exp7 and Exp8. From Exp4 on, each 2Wiki experiment excludes the questions used in the earlier experiments. The second corpus is a movie-domain subset of T-REx [30] with 482 documents and 14 relation types, of which 386 documents and 507 questions are held out for evaluation. Exp1 uses a bank of building-ontology questions and appears only in Appendix A. Exp7 holds 480 held-out questions in 30 shards and two run seeds, with no failed cells. On T-REx, the Exp7 arms use Qwen on 33 held-out shards with one-hop gold. Two lists were used there: the T-REx arms of the

TABLE 1
The arms compared. "Reads and writes" gives what each arm takes in and produces; the last column gives where the relation list enters the prompts when the arm runs with it.

Arm	Reads and writes	CQs	Where the list enters
L0 (single pass)	Chunks → 40 CQs → schema of at most 36 properties → facts extracted from every chunk	yes	CQ generation and schema derivation
G (Exp4, Exp5)	L0 with the CQ set expanded until every chunk is a CQ source	yes	as L0
GR (Exp4, Exp5)	G plus evaluate-and-repair rounds	yes	as L0
LOOP	L0, then three rounds of CQ evaluation and repair with 20 new CQs per round	yes	CQ generation, schema extension and schema patches
Combined	Relation list, CQs expanded to every chunk, an incremental schema with deterministic de-duplication, and extra extraction passes up to the token total of LOOP+vocab on the same unit; CQ evaluation only as an acceptance test, no repair step. "First pass only" stops after the first full extraction pass	yes	CQ generation, schema derivation and schema extension
U-ext (extraction without CQs)	No CQs and no schema: free-form predicates from every chunk, one call per chunk; the schema is derived afterwards	no	Extraction prompt, after the chunk text and before the instruction (U-ext+vocab)
+vocab	Any arm above run with the 22-name relation list (L0+vocab, LOOP+vocab, and U-ext+vocab, called *seeded extraction*; "Combined" always has it unless marked no list)	as arm	as in the arm's row
L0+vocab$K$	L0 given K of the 22 names (K = 6, 11, 17; nested random subsets)	yes	CQ generation and schema derivation
iText2KG, EDC, AutoSchemaKG	Chunks → facts with each public extractor's released defaults and no relation list (EDC also once with the list, Section 5); EDC and AutoSchemaKG induce and canonicalise their own schema, and AutoSchemaKG's entity-relation triples are scored	no	none

Exp7 comparisons (Table 12, Fig. 3) used the 2Wiki relation names, and the arms of Table 4 used the list induced from the T-REx corpus. The 2Wiki checks and Table 4 use two run seeds; the T-REx Exp7 arms and the AutoSchemaKG arm use one.

Metrics: Evidence recall and all QA measures are scored by code against gold. An LLM judge is used in three places. The builder's own model family (Qwen3.8-Flash-Next) judges the pipeline's internal CQ evaluation (stage 4) and the pass rate of fresh CQs, and a model from the other family judges precision in Exp8. Gemma judges Qwen graphs and Qwen judges Gemma graphs, and the CQ checks use the builder's family.

- *Evidence recall* is the share of gold evidence triples present in the graph. A triple counts as present when a chunk-cited triple links the same entity pair under a predicate of the same relation class. *Uncapped recall* counts every such triple. *Capped recall* is recall under a per-question triple budget: the budget is two triples for every source sentence of the question's paragraphs, and each gold evidence counts the probability that a uniform random subset of that question's graph slice no larger than the budget still holds a matching triple, computed exactly as a hypergeometric expectation without sampling, so padding the graph lowers recall. "Token budget" always refers to tokens.
- *Answer correctness* is scored by code against the gold answer without the grounding requirement: normalised exact match against the gold answer and its aliases or demonyms, the first token for yes/no questions, exact match for dates, and "unknown" counts as wrong.
- *Grounded QA* asks a model to answer each held-out question from facts retrieved from the graph. Given an input query q and graph G, let $R(q, G) \subseteq G$ denote the retrieved supporting fact subgraph. The grounded correctness indicator $\mathbb{I}_{\text{gc}}(q) \in \{0, 1\}$ evaluates whether the answer generated from $R(q, G)$ is correct and grounded in the retrieved facts. Grounded QA over a set Q of held-out questions is $\frac{1}{|Q|} \sum_{q \in Q} \mathbb{I}_{\text{gc}}(q)$. For comparison questions every gold evidence pair (subject and object, in either direction) must appear in a retrieved fact that cites a chunk containing the evidence. For open questions the gold answer must appear as the subject or object of a retrieved fact. The relation name plays no part in grounded QA; relation classes enter only evidence recall (class-aligned recall). QA-time retrieval is the same for every arm and does not use the list: the question is linked to entity labels (with aliases) by exact

TABLE 2
The eight studies (Exp6 is an error analysis, the others each vary one component), the Gemma replication and a third model. Each has a plan, a deviation log, code and results in the released archive.

	Question	Data	Models
Exp1	Does iterative CQ refinement improve a graph?	Building ontology bank, 35 test questions	Qwen
Exp2	Does iteration improve on one batch of the same CQs?	2Wiki, 24 shards	Qwen
Exp3	Does prompt optimisation improve on a relation list?	2Wiki, 400 test questions	Qwen
Exp4	Does a CQ $\rightarrow$ SPARQL $\rightarrow$ judge $\rightarrow$ patch loop help?	2Wiki, 40 shards	Qwen
Exp5	Does a loop with query repair and a rejecting gate help?	2Wiki, 24 shards $\times$ 2 run seeds	Qwen
Exp6	Which error types occur, and can Shapes Constraint Language (SHACL) see them?	Exp2 graphs	—
Exp7	What do the combined pipeline's parts add?	2Wiki, 30 shards $\times$ 2 run seeds	Qwen
Exp8	How do CQ count, budget and hybrid QA interact?	2Wiki and T-REx, 20 to 33 shards	Qwen, Gemma
Replication	Gemma (Exp1, Exp4, Exp5, Exp7, and Exp7's design on T-REx); Exp2 and Exp3 were not repeated	as above	Gemma
Third model	Nemotron (Exp5, Exp7 and T-REx; Appendix Table 11)	as above	Nemotron

n-gram matching, the graph is followed for up to two hops, each entity contributes at most 25 facts ranked by BM25, and the facts in $R(q, G)$ are ordered by hop, then BM25, and capped at 60.

- The *pass rate of fresh CQs* (Exp7) uses 40 new CQs per shard and run seed. They are written from 40 random chunks with a neutral prompt and no relation list, and no loop uses them.
- For each final graph, SPARQL is generated three times for each fresh CQ. A query passes when it returns at most 20 rows, the expected answer span is among them and the judge scores it 2. The pass rate is the mean over the three generations.
- *Hybrid QA* gives the answerer up to 30 graph facts (main run; a rerun used up to 60) and the top 3 retrieved passages, and *vector QA* gives it the top 5 passages. Exp8's KG-only QA and its measure F2 are defined in Appendix A.8.
- *Precision* is judged on 40 sampled facts per graph by the cross-family judge. Gold evidence covers only the facts that questions need, so unaligned triples are not counted as errors. In the fact precision check on the Exp7 graphs, Gemma judges Qwen facts in one pass.

Models: We implemented this family of CQ-driven methods as one pipeline. Construction, querying and QA use Qwen3.8-Flash-Next (NVFP4 weights) in Exp1–Exp7, and the Gemma replication repeats the experiments named in Table 2 with Gemma-4-26B-A4B-it. NVIDIA Nemotron-3.5-Lightning-30B-A3B, a third model, ran Exp5, Exp7 and T-REx; its graphs stayed at or below .047 grounded QA in every arm, so model comparisons rest on Qwen and Gemma. Exp8 runs Gemma-4-26B-A4B-it and Qwen3.8-Flash-Next on both corpora. All models were served with vLLM, and Qwen ran with thinking off. The QA answerer decodes at temperature 0 (120 output tokens) and extraction at temperature 0 (1,500 output tokens). First-round CQ generation uses temperature 0.7, later CQ steps use 0, and the judges use 0.

External extractors: As an external reference, iText2KG 1.1.0 ran with its default settings on each chunk of the Exp7 2Wiki shards with Qwen. It takes no relation list, so none was given, and the Exp7 answerer and scorer scored its graphs. The public extractor EDC [28] ran on the same chunks, model and engine with its released defaults (Self Canonicalization). It induces and canonicalises its own schema, so it took no list. A third public extractor, AutoSchemaKG [11], ran with its released defaults on the same chunks, model family, answerer and scorer, on 30 shards with one run seed; it also took no list.

Registration and statistics: Each experiment has a plan, fixed before the main run, that states hypotheses and decision rules; the labels, the checks added later and the run records are in Appendix A.9. Status labels follow the plan: *registered* hypotheses were fixed beforehand, *secondary* measures are reported beside them, *descriptive* contrasts report intervals only, and *post hoc* analyses and *added arms* came after the results or after the plan was frozen. The unit of analysis is the shard in Exp2, Exp4, Exp5, Exp7 and Exp8 and the question in Exp1 and Exp3; within a shard we average values over run seeds and compare arms pairwise. Intervals are two-sided 95% bias-corrected and accelerated (BCa) bootstrap intervals that resample by shard, with 10,000 resamples in Exp7 and Exp8. Primary hypotheses are Holm-corrected within the registered primary family of each plan. Tables give the number of shards as n and state units where both appear (30 shards are 60 shard $\times$ run seed units).

Chunk retrieval: On the same 2Wiki units of Exp7, two arms answer from the top 5 chunks retrieved by em-

bedding. Top 5 matches the graph arm's prompt length (791 against 794 tokens, median); one arm also receives the 22-name list as a hint. Answer correctness is the measure, since grounding is defined on graph facts.

Controls for volume: Tokens per unit count the construction of an arm and its own CQ evaluation: extraction, CQ generation, the pipeline's SPARQL and judge calls and repair, with retries counted and cached calls billed at their nominal cost. They exclude the held-out QA that scores the graph. The construction of an induced list is a separate step and is not included. Most comparisons hold volume fixed in one of three ways. Arms with equal token budgets stop at the same share of a reference budget (Exp7, Exp8), and the combined pipeline's budget was the token total of LOOP with the list on the same unit. Its arm without the list used the same budget (837,655 and 838,162 tokens). Exp4 and Exp5 compare evaluation-guided patches with random patches of the same size. In Exp7 we also match fact counts. The check at equal fact count draws a uniform random subset of the facts in each stored U-ext graph, with and without the list. The subset matches the fact count $|E|$ of the corresponding L0 graph, and the same answerer and scorer score it. Its plan was fixed before any model call. Two selections reduced the same graphs to the same fact count. One keeps the facts whose relation is most frequent, the other the facts whose end entities are most central. Both read only the facts already in the graph. A chain-preserving and a model selector read no questions, gold or CQs either and keep exactly the guided pass's fact count: the first keeps the facts that add the most connected two-hop entity paths, and the second has the answer model score each fact's informativeness from its source chunk alone. Two more selectors, over the same stored graph and fact count, read no questions and no gold either: a selection by the CQs (BM25 between each fact and the 40 CQs that the guided pass generated for that unit; top facts kept) and a chunk-balanced selection (facts taken round-robin by chunk, ties by hash).

4 Relation Vocabulary Raises Held-Out QA

The relation list raised held-out grounded QA in all four Exp7 pipelines of Qwen on 2Wiki (Table 7, Fig. 2). L0 gained 14.7 points ([10.9, 18.4]), LOOP 9.4, the combined pipeline 9.3 and U-ext 4.5, a range of 4.5 to 14.7. The arms are defined in Table 1. In Gemma on 2Wiki the list raised grounded QA over no list by 7.4 points in L0 ([3.3, 11.3]) and 7.5 in LOOP ([1.3, 15.0]). The Gemma combined pipeline gained 9.3 ([5.7, 12.9]). Table 3 lists the main contrasts and their status.

Synonym names kept the benefit in L0. No question ID or normalised question text of the list's training split appears among the evaluation questions, and no synonym shares a word stem with its original name. In L0 answer correctness rose by 15.4 points (Qwen, 2WikiMultihopQA; [11.3, 19.5]; Table 6), and grounded QA rose by 15.8 ([11.5, 20.3]). In U-ext, which already extracts from every chunk, the synonym list was on par with no list (+1.6, [−1.4, 4.2]), in line with the smaller effect of the original list there (Appendix Table 14).

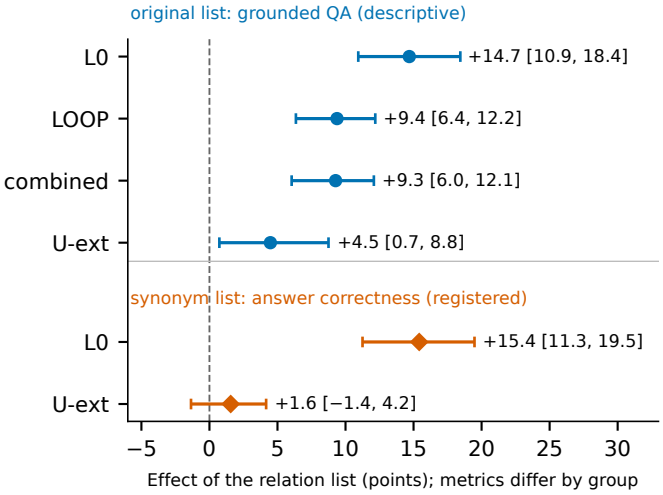


Fig. 2. The relation list raised held-out grounded QA in all four pipelines, and synonym names raised answer correctness in L0. Differences in points with 95% intervals, Qwen on 2Wiki. The two groups share one axis but use different metrics, so compare points within a group only. Top group, original list: grounded QA. Bottom group, synonym list: answer correctness; L0 +15.4 [11.3, 19.5], U-ext +1.6 [−1.4, 4.2].

In L0 the synonym list was on par with the original list in answer correctness (+1.7, [−2.0, 5.0]); the original list gained 13.8 over no list in the stored runs.

A list induced from the corpus, with no gold relations, raised L0 answer correctness by 16.3 points ([11.6, 20.2]). The original list rerun beside it gave +15.1 ([10.5, 19.5]).

The same induction on T-REx gave a list of 22 names, again with no question and no gold relation. It raised the single pass by 13.3 points of grounded QA ([8.0, 23.1]) and by 12.5 points of answer correctness (Table 4).

A second T-REx list, induced from 96 passages outside the evaluation shards with the same procedure and no question, shares 15 of its 22 names with the first and 6 with the 2Wiki list. With it, the single pass scored 7.4 points above no list in grounded QA ([3.9, 11.4]; Table 4), the same pattern as the first T-REx list.

A second 2Wiki list, induced with the same procedure from 300 of 6,136 chunks outside the evaluation shards and no question, shares 18 of its 22 names with the first induced list and 5 with the training-split list. With it, the single pass scored 15.8 points above no list in grounded QA ([9.6, 21.9]; Table 13). Extraction from every chunk gained 3.1 points in grounded QA ([0.0, 6.5]) and was on par with no list in answer correctness (+1.7, [−1.0, 4.2]).

Answer correctness in L0 depended on how much of the list was given (Table 5). The curve is not monotone. Eleven of the 22 names scored 10.5 points above no list ([5.5, 14.5]), while 6 names scored below no list. The dip at 6 of 22 names is consistent with narrowing: 51.4% of the extracted facts carried those six names against 6.5% without a list. With the full list (Section 3.1), which covers 98.4% of the gold evidence, the score rose to .568 from .417.

TABLE 3
Results at a glance. Values are differences in points unless marked; n is the number of shards; where two run seeds were used, 30 shards are 60 shard × run seed units. Status is the contrast's standing in the plan. A dash means the contrast was not repeated. For the equal-fact-count rows, the registered test asks whether the selection scores above the guided pass, and the status reports that it did not.

Claim	Exp	Model, corpus	Contrast	Value [95% interval]	n	Status
The list raises grounded QA	Exp7	Qwen, 2Wiki	With vs. without list, four pipelines	L0 +14.7 [10.9, 18.4]; LOOP +9.4 [6.4, 12.2]; combined +9.3 [6.0, 12.1]; U-ext +4.5 [0.7, 8.8]	30	Descriptive (intervals only)
Synonym list, answer correctness	Exp7	Qwen, 2Wiki	Synonym list − no list	L0 +15.4 [11.3, 19.5]; U-ext +1.6 [−1.4, 4.2]	30	Registered: improved (L0); not established (U-ext)
Induced list raises answer correctness	Exp7	Qwen, 2Wiki; T-REx	Induced list − no list, L0	2Wiki +16.3 [11.6, 20.2]; T-REx +12.5 (Table 4)	30; 33	Registered: improved
List induced outside the evaluation shards raises grounded QA	Exp7	Qwen, 2Wiki; T-REx	Outside-shard list − no list, L0	2Wiki +15.8 [9.6, 21.9]; T-REx +7.4 [3.9, 11.4]	30; 33	Registered: improved
Graph above chunk retrieval, answer correctness	Exp7	Qwen, 2Wiki	U-ext+vocab − retrieval of top 5 chunks; with list hint	+15.4 [10.6, 19.8]; +16.9 [11.1, 22.3]	30	Registered: improved
Public extractors without a list below seeded extraction, grounded QA	Exp7	Qwen, 2Wiki	iText2KG, EDC, AutoSchemaKG − seeded extraction	−17.3 [−21.0, −13.9]; −15.8 [−19.3, −12.3]; −39.2 [−44.0, −34.4]; U-ext without list − EDC +11.4 [7.5, 15.2]	30	Descriptive
Equal fact count: guided single pass above random subset of U-ext	Exp7	Qwen, 2Wiki	L0 − random subset of U-ext at L0's fact count; with list, L0+vocab − random subset of U-ext+vocab, QA	+28.2 [23.4, 33.2]; +39.1 [34.2, 44.1]	30	Registered: random subset not above the guided pass (both)
Expansion raises grounded QA	Exp4, Exp5	Qwen, 2Wiki	G − L0	+9.1 [4.7, 13.8]; +9.6 [5.2, 14.8]	40; 24	Post hoc secondary
Iteration did not separate from one batch of the same CQs	Exp2	Qwen, 2Wiki	Iteration − one batch, recall	+1.2 [−3.5, 5.7], crossed shard × run seed interval	24	Registered primary (inconclusive)
Hybrid QA: no detectable change at 1×, lower at 4× budget in Qwen on 2Wiki	Exp8	as above	160 vs. 40 CQs, 1×	−3.6 to +0.6, all intervals include 0 (Table 10). At 4× budget, Qwen 2Wiki −9.1 [−13.8, −5.5]	20–33	Registered primary
More CQs raise KG-only QA	Exp8	Qwen, Gemma; 2Wiki, T-REx	160 vs. 40 CQs, 1×	+3.5 to +10.9, all intervals above 0 (Table 10)	20–33	Secondary measure
Guided repair not improved over random	Exp4, Exp5	Qwen, 2Wiki	Guided − random patches, QA	+1.9 [−1.6, 5.3]; −0.3, one-sided upper bound 3.1	40; 24	Exp5 primary, margin 5; Exp4 inconclusive
Combined pipeline leads loop on uncapped recall and recall at equal fact counts, trails on capped recall	Exp7	Qwen, 2Wiki	Combined − LOOP+vocab, recall	Uncapped +5.6 [4.2, 7.0]; capped −8.7 [−10.6, −6.8]; equal fact counts +1.1 [0.5, 1.9]	30	Registered primary (uncapped); matching in plan
Seeded extraction without CQs was not distinguishable from the combined pipeline	Exp7	Qwen, 2Wiki	U-ext+vocab − Combined, QA	−0.3 [−3.5, 3.1]; 106k per unit against 277k (first pass) and 838k (full run); answer correctness +2.3 [−0.2, 4.4] (post hoc)	30	Added arm
Combined pipeline passes more fresh CQs than seeded extraction without CQs	Exp7	Qwen, 2Wiki	Combined − U-ext+vocab, fresh-CQ pass rate	+13.5 [10.7, 16.3]; LLM judge of the builder's family	30	Secondary, in the Exp7 plan; added arm
Most inexact gold facts are omissions	Exp6	Qwen, 2Wiki	Omission share of inexact gold facts	87.3% of 1,336	—	Analysis of Exp2 graphs

TABLE 4
The lists induced for T-REx (Qwen, 33 shards; the last two rows use the list induced outside the evaluation shards, one run seed), differences in points with 95% intervals. Reg.: registered; Desc.: descriptive. Run seeds 1 and 2 use the same 33 shards. The Desc. row compares extraction with the induced list against the stored combined-pipeline run on T-REx, which used the original list.

	Grounded QA	Answer correct	Status
Single pass, with list − without list			
Run seed 1	+13.3 [8.0, 23.1]	+12.5	Reg.: improved
Run seed 2	+14.1 [9.1, 23.4]	+13.5	Reg.: improved
No CQs, with list − without list			
Run seed 1	+0.5 [−2.5, 2.9]	—	Reg.: not established
Run seed 2	+0.6 [−2.7, 4.1]	−0.3	Reg.: not established
Induced-list extraction − stored combined run			
Run seed 1	+5.8 [0.8, 10.3]	+7.6 [3.0, 12.6]	Desc.
Outside-shard list − no list			
Single pass	+7.4 [3.9, 11.4]	+6.8 [3.0, 10.9]	Reg.: improved
No CQs	−2.4 [−5.2, 0.4]	−1.8 [−4.7, 1.1]	Reg.: not improved

TABLE 5
Mean answer correctness in L0 by list content, with 95% shard-cluster BCa intervals (Qwen on 2Wiki, Exp7 design, 60 units). Names are the share of the 22 original names given, in nested random subsets, each partial setting one draw of names. Values are reruns from one serving instance (stored Exp7 values: no list .433, original list .571).

Relation list	Answer correct [95% interval]
No list	.417 [.366, .474]
6 of 22 names	.325 [.274, .383]
11 of 22 names	.522 [.481, .554]
17 of 22 names	.550 [.500, .592]
22 original names	.568 [.521, .613]
22 induced names	.579 [.533, .620]

5 Seeded Extraction Reaches the Combined Pipeline's Grounded QA With Fewer Tokens

With the same relation list, extraction from every chunk without CQs reached .603 grounded QA against the combined pipeline's .606, on par (−0.3 points, [−3.5, 3.1]), and used 106k tokens per unit. The combined pipeline's first pass alone, without its extra extraction passes, reached .607 grounded QA at 277k tokens per unit, and its full run .606 at 838k (Table 7, Fig. 3). On answer correctness seeded

TABLE 6
Synonym-list check, differences in points (95% BCa intervals, 30 shards, 60 units, Qwen on 2Wiki). Class-free recall ignores relation classes; class-aligned recall is the scorer's primary recall.

	Synonym list − no list		Synonym − original list	
	Answer correct	Class-free recall	Answer correct	Class-aligned recall
L0	+15.4 [11.3, 19.5]	+12.4 [9.2, 16.0]	+1.7 [−2.0, 5.0]	−13.7 [−19.6, −8.4]
U-ext	+1.6 [−1.4, 4.2]	+0.4 [−0.5, 1.5]	−1.5 [−4.2, 1.4]	−10.1 [−11.8, −8.8]

TABLE 7
Exp7 arm means on 2WikiMultihopQA (Qwen). "Combined, first pass only" is the combined pipeline without its extra extraction passes. QA is held-out grounded QA. Rec. unc. and Rec. cap. are uncapped and capped evidence recall. Facts and tokens (Tok.) are per shard and run seed, tokens in thousands. The combined pipeline without the list runs at the same token budget as the combined pipeline, hence 838k for both. Differences in the text are computed from unrounded means.

	QA	Rec. unc.	Rec. cap.	Facts	Tok. (k)
L0	.394	.646	.609	725	131
L0+vocab	.541	.761	.726	687	108
LOOP	.505	.768	.618	1,134	1,061
LOOP+vocab	.599	.845	.731	980	878
U-ext	.558	.874	.512	1,704	100
U-ext+vocab	.603	.895	.604	1,472	106
Combined	.606	.901	.643	1,435	838
Combined, no list	.514	.798	.562	1,418	838
Combined, first pass only	.607	.854	.742	972	277

extraction was on par as well (+2.3, [−0.2, 4.4]). It wrote more facts than L0+vocab (1,472 against 687).

On T-REx, with every arm given the original 22 names, seeded extraction scored .421 grounded QA against .371 for the combined pipeline, 5.0 points higher (90% interval [1.1, 8.1]), at 19.1k tokens per unit against 298k.

Three public extractors in the same setup: None of the three public extractors received a relation list, and each ran with released defaults. Extraction without CQs and without a list (U-ext) scored .558 grounded QA on the same units (Table 7). Run with Qwen on 2Wiki in the Exp7 setup (60 units, none failed), iText2KG wrote 1,899 facts, more than any arm in Table 7, and reached .430 grounded QA (Table 8). That is 17.3 points below seeded extraction and 17.6 below the combined pipeline, at 478.8k tokens per unit against 105.9k and 837.7k. Against the single pass it was on par in grounded QA (+3.7, [−1.5, 9.3]) and 11.6 points higher in answer correctness. EDC, a second public extractor, induces its schema from the text and canonicalises it itself, and we ran it on the same model and engine with its released defaults (Self Canonicalization) [28]. It scored .445 grounded QA and .629 answer correctness (Table 8), 5.1 points ([0.5, 10.0]) and 19.6 points ([14.8, 24.7]) above the single pass. EDC scored 15.8 points below seeded extraction in grounded QA ([−19.3, −12.3]), at 618.4k tokens per unit against 105.9k. Given the same 22-name list as its target schema, with the names as definitions (its Target Alignment mode), EDC scored .458 grounded QA with the list against .454 without it (+0.4, [−2.7, 2.9]), with 824 against 1,356 facts per unit. Seeded extraction scored 14.6 points above EDC with the list ([10.4, 18.5]), so giving EDC the same list did not close the gap.

AutoSchemaKG, a third public extractor, also induces its schema from the text [11]. We ran it with its released defaults on the same model family, chunks, answerer and scorer, on 30 2Wiki shards with one run seed. Its entity-relation triples, the variant comparable to the other arms, scored .213 grounded QA and .446 answer correctness, with 1,256.6 facts at 1,557.8k tokens per unit (Table 8). It scored 39.2 points below seeded extraction in grounded QA ([−44.0, −34.4]) and 41.9 below the combined pipeline ([−46.3, −38.3]). Against the single pass it scored 19.2 points lower in grounded QA ([−26.0, −13.1]) and was on par in answer correctness (+2.1, [−5.0, 9.4]).

The gold evidence pairs reached the 60 retrieved facts in 88.5% of cases for seeded extraction, 81.0% for iText2KG, 75.6% for EDC and 57.7% for AutoSchemaKG triples. Requiring a matching relation class changed each arm's grounded QA by at most 1.2 points and left the gaps unchanged or wider, and relaxing entity linking changed these reach rates by at most 0.5 points in every arm.

Retrieval over the same chunks: On the same 2Wiki units, seeded extraction reached .669 answer correctness in a rerun beside the retrieval arms (.678 in the stored run, Table 8). Retrieval-augmented generation from the top 5 chunks reached .515, and .500 with the 22-name list as a hint. Answer correctness is the common measure because the retrieval arms have no graph facts, so grounded QA cannot be computed for them. The graph scored 15.4 points above chunk retrieval ([10.6, 19.8]) and 16.9 above list-hinted retrieval ([11.1, 22.3]). With no context, the answer model reached .325 from its own knowledge, 35.3 points below the stored seeded run (.678; [30.1, 40.1]).

The grounded-QA results above use graph facts only. When passages were added to the retrieved facts (hybrid QA) and the graph facts were capped at 60, every pipeline scored above vector-only QA (.502): LOOP+vocab .725, L0+vocab .716, seeded extraction .709, Combined .699 and L0 .626 (60 units). Raising the fact cap from 30 to 60 raised answer correctness by 12.4 points in seeded extraction ([9.3, 15.3]) and 9.6 in Combined ([7.2, 11.7]). In the stored run with up to 30 facts the scores were L0+vocab .692, LOOP+vocab .671, Combined .603, L0 .600 and seeded extraction .585 (Appendix Table 12): with 30 facts the arms with the most facts ranked lowest, and with 60 the four list or full-extraction arms were close.

6 What CQs Add: Each Fact Counts

With the relation list and an equal fact count, the guided single pass scored 14.3 points above the model-scored selection ([8.9, 19.3]) and 25.0 above the chain-preserving selection ([19.8, 30.1]) of seeded extraction in grounded QA (Appendix Table 15). It scored 39.1 points above an equal-size random subset ([34.2, 44.1]), and the random subset did

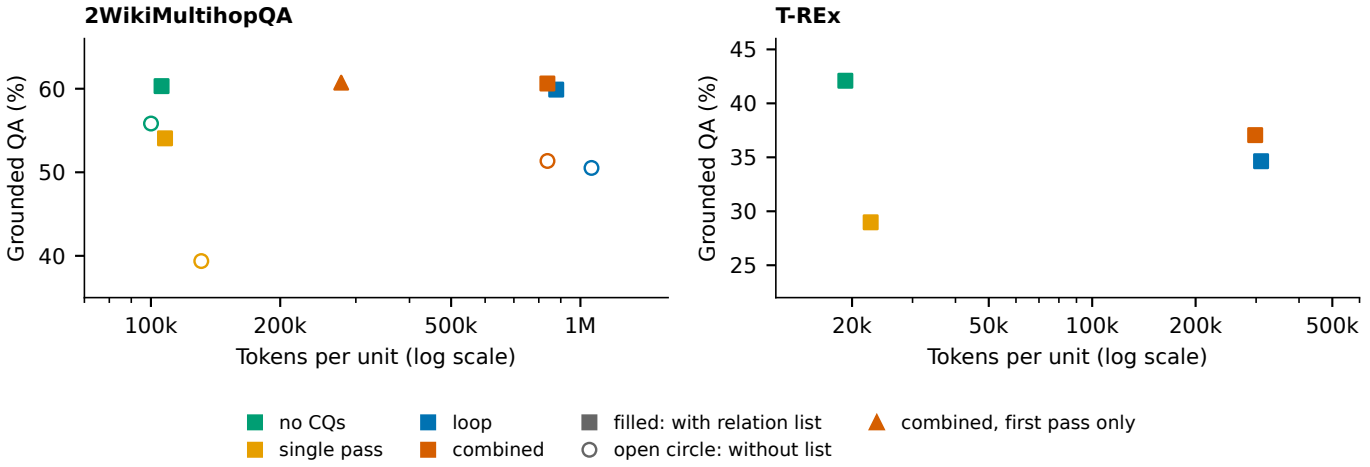


Fig. 3. Seeded extraction without CQs reached the combined pipeline's grounded QA at a fraction of its tokens (106k per unit, against 277k for the combined pipeline's first pass and 838k for its full run). Points are arm means of grounded QA against tokens per unit (log axis); intervals for the contrasts are in Tables 3 and 8. Left, 2Wiki: arms with and without the list. Right, T-REx: seeded arms only. Qwen.

TABLE 8
Three public extractors against Exp7 arms (Qwen, 2Wiki, 60 shard × run seed units). EDC ran with its released defaults (Self Canonicalization). The AutoSchemaKG rows cover 30 shards with one run seed, and their differences are taken against the stored arms on those shards (grounded QA, answer correctness: seeded extraction .604, .679; combined .631, .677; single pass .404, .425). The EDC means with and without the list in the text (.458, .454) are on 30 units of one run seed and differ from the 60-unit values here. AutoSchemaKG's entity-relation triples are the comparable variant; its full graph also holds concept and event edges. Tokens are per unit, in thousands. The difference rows give each extractor minus the arm, in points, with 95% intervals.

	Grounded QA	Answer correctness	Facts	Tokens (k)
iText2KG	.430 [.393, .470]	.549 [.503, .596]	1,899	478.8
EDC	.445 [.398, .493]	.629 [.583, .676]	1,357.7	618.4
AutoSchemaKG, entity-relation triples	.213 [.175, .252]	.446 [.396, .490]	1,256.6	1,557.8
AutoSchemaKG, full graph	.096 [.073, .121]	.233 [.200, .260]	18,360	1,557.8
Seeded extraction	.603	.678	1,472	105.9
Combined	.606	.655	1,435	837.7
Single pass (L0)	.394	.433	725	131.0
iText2KG − seeded extraction	−17.3 [−21.0, −13.9]	−12.9 [−16.3, −10.0]		
iText2KG − combined	−17.6 [−20.8, −14.1]	−10.6 [−14.1, −7.5]		
iText2KG − single pass	+3.7 [−1.5, 9.3]	+11.6 [5.6, 18.2]		
EDC − seeded extraction	−15.8 [−19.3, −12.3]	−4.9 [−7.7, −2.5]		
EDC − combined	−16.1 [−20.1, −12.4]	−2.6 [−5.9, 0.5]		
EDC − single pass	+5.1 [0.5, 10.0]	+19.6 [14.8, 24.7]		
AutoSchemaKG triples − seeded extraction	−39.2 [−44.0, −34.4]	−23.3 [−27.5, −20.0]		
AutoSchemaKG triples − combined	−41.9 [−46.3, −38.3]	−23.1 [−27.3, −19.0]		
AutoSchemaKG triples − single pass	−19.2 [−26.0, −13.1]	+2.1 [−5.0, 9.4]		

not score higher. The lead held against selections made with the CQs or balanced over chunks (below). Without the list the relation-frequency selection scored 8.0 points above the guided pass ([2.7, 13.3]), and on T-REx without the list the guided pass and a random subset were on par (6.6 points, [−5.8, 12.2]).

Without the list, U-ext raised grounded QA over L0 by 16.5 points ([10.6, 21.7]) by writing 1,704 facts against 725. The check compares the CQ-guided facts with facts chosen at random: we drew a uniform random subset of the stored U-ext graph (or of the listed graph, for the listed comparison) down to the fact count of the matching single pass, and the same answerer and scorer scored it. Without the list, the single pass scored 28.2 points above an equal-size random subset of open extraction ([23.4, 33.2]). On T-REx with the list, the single pass scored 7.4 points above an equal-size random subset of seeded extraction ([2.9, 12.8]). Two heuristic selections, one by relation frequency and one by entity centrality, cut the extraction graph to the guided graph's fact count. With the list, the CQ-guided single pass stayed ahead of both, by 16.6 points against the relation-frequency selection ([11.4, 22.1]) and 25.8 against the centrality selection ([20.5, 30.5]) in grounded QA (Appendix Table 15).

On 2Wiki with the list (60 units), the guided pass scored 25.0 points above the chain-preserving selector ([19.8, 30.1]) and 14.3 above the model selector ([8.9, 19.3]) in grounded QA (Appendix Table 15). Selections by the CQs of the unit and balanced over chunks scored below the guided pass by 33.3 points ([28.3, 37.7]) and 37.2 points ([31.6, 41.4]). This separates CQ-guided extraction from CQ-guided selection.

Structured queries from fresh CQs: CQs also leave a schema that answers structured queries. On 40 fresh CQs per shard and run seed, written from random chunks with

a neutral prompt, the combined pipeline passed 13.5 points more through SPARQL than seeded extraction ([10.7, 16.3]). Held-out QA is reached by the simpler baseline, and on questions asked as CQs the CQ pipeline leads.

7 Discussion

A relation list lifts extraction: The relation list raised grounded QA in every Exp7 pipeline on 2Wiki. Neither gold relations nor the exact names were needed, since synonyms, a corpus-induced list and lists induced outside the evaluation shards raised the single pass on both corpora (Section 4). The full list and the induced lists, which cover most gold relations, raised every pipeline.

Seeded extraction against the combined pipeline and the public extractors: Seeded extraction reached the combined pipeline's grounded QA (Section 5), and its fact precision passed the non-inferiority test (.925 against .716). The public extractors received no relation list, except in one run with the list (Section 5). iText2KG wrote the most facts of any arm yet scored below seeded extraction (Table 8), so writing more facts did not by itself deliver that score. On 2Wiki the listed graph answered more questions correctly than retrieval over the same chunks, while the list given to retrieval as a hint did not raise answer correctness.

At a fixed fact count, CQ-guided facts scored higher: At equal fact count and with the relation list, the CQ-guided single pass scored above a uniform random subset of seeded extraction, on the multi-hop corpus (Section 6) and on T-REx. It also scored above two heuristic subsets of the same graph at the same fact count, one keeping the facts of the most frequent relations and one the facts whose end entities are most central, and above a chain-preserving and a model-scored selector (Appendix Table 15).

8 Conclusion

In this pipeline, how well a graph answered held-out questions followed what reached it. A relation list induced from the corpus, without gold relations, raised answer correctness by 16.3 points on 2WikiMultihopQA and 12.5 on T-REx, and lists induced only outside the evaluation documents also raised grounded QA on both corpora. Once the extractor had a relation list, extraction from every chunk without CQs reached the combined pipeline's grounded QA on the multi-hop corpus, at 106k tokens per unit against 277k for the combined pipeline's first pass and 838k for its full run. Without any list it scored .558 grounded QA against .430 for iText2KG, .445 for EDC and .213 for AutoSchemaKG. On fresh CQs run through SPARQL, the combined pipeline passed 13.5 points more queries than seeded extraction: a simple baseline reaches the combined pipeline's held-out QA, while the CQ pipeline leads on structured queries asked through CQs. At equal fact count, the CQ-guided single pass scored 14.3 and 25.0 points above model-scored and chain-preserving selections of the listed extraction, and above a random subset and the heuristic, CQ-based and chunk-balanced selections. A corpus-induced relation vocabulary with extraction from every chunk is a simple, strong baseline for CQ-driven graph construction, and with a fixed fact count CQ guidance kept more answerable facts than random and question-agnostic subsets.

References

[1] M. Grüninger and M. S. Fox, "Methodology for the design and evaluation of ontologies," in *Workshop on Basic Ontological Issues in Knowledge Sharing, IJCAI-95*, Montreal, 1995.

[2] N. F. Noy and D. L. McGuinness, "Ontology development 101: A guide to creating your first ontology," Knowledge Systems Laboratory, Stanford University, Technical Report KSL-01-05, Mar. 2001.

[3] X. Ho, A.-K. D. Nguyen, S. Sugawara, and A. Aizawa, "Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps," in *Proceedings of the 28th International Conference on Computational Linguistics*. Barcelona, Spain (Online): International Committee on Computational Linguistics, 2020, pp. 6609–6625.

[4] X. Feng, X. Wu, and H. Meng, "Ontology-grounded automatic knowledge graph construction by LLM under wikidata schema," 2024, presented at HI-AI@KDD, KDD 2024; CEUR Workshop Proceedings 3841 (2024) 117–135.

[5] A. Kaiser, C. Leoveanu-Condrei, R. Gold, M.-C. Dinu, and M. Hofmarcher, "HyDRA: A hybrid-driven reasoning architecture for verifiable knowledge graphs," 2025.

[6] S. Choi, W. Jeon, K. Yang, and T. Kim, "SocraticKG: Knowledge graph construction via QA-driven fact extraction," 2026.

[7] M. Huang, C. Bu, Y. He, and X. Wu, "How to mitigate information loss in knowledge graphs for GraphRAG: Leveraging triple context restoration and query-driven feedback," 2025, accepted to IJCAI 2025.

[8] M. Chong, S. Zhang, J. Fan, and X. Du, "EvoOntology: A self-evolving ontology layer for data agents," 2026.

[9] X. Zhang, Z. Sun, C. Yang, Y. Cui, L. Guo, and W. Hu, "Toward effective and reliable LLM agents via dynamic ontology," 2026.

[10] O. Sainz, I. García-Ferrero, R. Agerri, O. Lopez de Lacalle, G. Rigau, and E. Agirre, "GoLLIE: Annotation guidelines improve zero-shot information-extraction," in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024, arXiv:2310.03668.

[11] J. Bai, W. Fan, Q. Hu, Q. Zong, C. Li, H. T. Tsang, H. Luo, Y. Yim, H. Huang, X. Zhou, F. Qin, T. Zheng, X. Peng, X. Yao, H. Yang, L. Wu, J. Yi, G. Zhang, R. Chen, and Y. Song, "AutoSchemaKG: Autonomous knowledge graph construction through dynamic schema induction from web-scale corpora," in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. San Diego, California, USA: Association for Computational Linguistics, 2026, pp. 20 557–20 584, arXiv:2505.23628.

[12] B. Zhang, V. A. Carriero, K. Schreiberhuber, S. Tsaneva, L. S. González, J. Kim, and J. de Berardinis, "OntoChat: a framework for conversational ontology engineering using language models," 2024, eSWC 2024 Special Track on Large Language Models for Knowledge Engineering.

[13] A. S. Lippolis, M. J. Saeedizade, R. Keskisärkkä, S. Zuppiroli, M. Ceriani, A. Gangemi, E. Blomqvist, and A. G. Nuzzolese, "Ontology generation using large language models," 2025.

[14] A. S. Lippolis, M. J. Saeedizade, R. Keskisärkkä, A. Gangemi, E. Blomqvist, and A. G. Nuzzolese, "Large language models assisting ontology evaluation," 2025.

[15] E. Watkiss-Leek, R. Alharbi, H. Rostron, A. Ng, E. Johnson, A. Mitchell, T. R. Payne, V. Tamma, and J. de Berardinis, "IDEA2: Expert-in-the-loop competency question elicitation for collaborative ontology engineering," 2026.

[16] A. Talukder, M. A. Mridul, and O. Seneviratne, "Towards automated ontology generation from unstructured text: A multi-agent LLM approach," 2026.

[17] O. Khattab, A. Singhvi, P. Maheshwari, Z. Zhang, K. Santhanam, S. Vardhamanan, S. Haq, A. Sharma, T. T. Joshi, H. Moazam, H. Miller, M. Zaharia, and C. Potts, "DSPy: Compiling declarative language model calls into self-improving pipelines," 2023.

[18] K. Opsahl-Ong, M. J. Ryan, J. Purtell, D. Broman, C. Potts, M. Zaharia, and O. Khattab, "Optimizing instructions and demonstrations for multi-stage language model programs," 2024, eMNLP 2024.

[19] L. A. Agrawal, S. Tan, D. Soylu, N. Ziems, R. Khare, K. Opsahl-Ong, A. Singhvi, H. Shandilya, M. J. Ryan, M. Jiang, C. Potts, K. Sen, A. G. Dimakis, I. Stoica, D. Klein, M. Zaharia, and O. Khattab, "GEPA: Reflective prompt evolution can outperform reinforcement learning," 2025, accepted to ICLR 2026 (Oral).

[20] N. Mihindukulasooriya, N. S. D'Souza, F. Chowdhury, and H. Samulowitz, "Automatic prompt optimization for knowledge

graph construction: Insights from an empirical study," 2025, accepted at LLM+Graph WS at VLDB 2025.
[21] H. T. Mai, R. Kochnev, C. X. Chu, L. Lange, H. Paulheim, and D. Stepanova, "APOLO: Automatic prompt optimization for ontology learning," 2026, accepted at the Posters and Demos Track of ISWC 2026.
[22] Y. Lairgi, L. Moncla, R. Cazabet, K. Benabdeslem, and P. Cléau, "iText2KG: Incremental knowledge graphs construction using large language models," 2024, accepted at the International Web Information Systems Engineering conference (WISE) 2024.
[23] B. Mo, K. Yu, J. Kazdan, J. Cabezas, P. Mpala, L. Yu, C. Cundy, C. Kanatsoulis, and S. Koyejo, "KGGen: Extracting knowledge graphs from plain text with language models," 2025.
[24] N. Mihindukulasooriya, S. Tiwari, C. F. Enguix, and K. Lata, "Text2KGBench: A benchmark for ontology-driven knowledge graph generation from text," 2023, accepted at ISWC 2023 (Resources Track).
[25] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, D. Metropolitansky, R. O. Ness, and J. Larson, "From local to global: A graph RAG approach to query-focused summarization," 2024.
[26] Z. Bi, J. Chen, Y. Jiang, F. Xiong, W. Guo, H. Chen, and N. Zhang, "CodeKGC: Code language model for generative knowledge graph construction," *ACM Transactions on Asian and Low-Resource Language Information Processing*, vol. 23, no. 3, 2024, arXiv:2304.09048.
[27] Z. Li, Y. Zeng, Y. Zuo, W. Ren, W. Liu, M. Su, Y. Guo, Y. Liu, X. Li, Z. Hu, L. Bai, W. Li, Y. Liu, P. Yang, X. Jin, J. Guo, and X. Cheng, "KnowCoder: Coding structured knowledge into LLMs for universal information extraction," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Bangkok, Thailand: Association for Computational Linguistics, 2024, pp. 8758–8779, arXiv:2403.07969.
[28] B. Zhang and H. Soh, "Extract, define, canonicalize: An LLM-based framework for knowledge graph construction," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Miami, Florida, USA: Association for Computational Linguistics, 2024, pp. 9820–9836.
[29] B. J. Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su, "HippoRAG: Neurobiologically inspired long-term memory for large language models," in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024, arXiv:2405.14831.
[30] H. Elsahar, P. Vougiouklis, A. Remaci, C. Gravier, J. Hare, F. Laforest, and E. Simperl, "T-REx: A large scale alignment of natural language with knowledge base triples," in *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*. Miyazaki, Japan: European Language Resources Association (ELRA), 2018.
[31] H. Knublauch and D. Kontokostas, "Shapes constraint language (SHACL)," W3C, W3C Recommendation, Jul. 2017. [Online]. Available: https://www.w3.org/TR/2017/REC-shacl-20170720/

APPENDIX A
ADDITIONAL RESULTS AND RECORDS

A.1 Further relation-list checks

Relation list names: Table 13 lists the 22 names of each 2Wiki list, copied from the list files, in alphabetical order.

Synonyms and alignment: Class-aligned recall fell under synonyms, as name alignment predicts. In U-ext the synonym list's answer correctness was −1.5 points against the original list ([−4.2, 1.4]). A post hoc re-analysis of stored predictions tests the original list's gain without name matching (Table 14). For the three pipelines that use CQs the effect stays above zero on both measures, and class-free recall keeps 65 to 85% of the class-aligned effect.

A.2 Seeded extraction against the combined pipeline

A post hoc 90% interval of [−3.0, 2.6] around the difference in grounded QA (Section 5) excludes differences beyond about 3 points in either direction. On T-REx the combined pipeline found more gold evidence (class-aligned recall +4.4, 90% [3.0, 6.0]).

A.3 Fact-count check and coverage

With the list, U-ext raised grounded QA over L0+vocab by 6.3 points ([2.6, 9.4], post hoc).

Heuristic selections at equal fact count: Appendix Table 15 gives the eight registered contrasts of the equal-fact-count check against the CQ-guided pass.

A.4 Hybrid QA and CQ count

Table 9 gives the CQ-count settings. On the Exp7 graphs (Qwen, 2Wiki), hybrid QA scored above vector-only QA (.502 [.460, .544]) in every pipeline (Table 12); the run with 60 graph facts is reported in Section 5. In Exp8, four times as many CQs at the same budget raised KG-only QA in all four settings (Table 10); hybrid QA, the registered primary measure, did not separate from the 40-CQ reference.

A.5 Evaluation-guided repair

Iterative CQ generation in Exp2 grew the CQ set from 12.0 to 85.8 per shard and evidence recall from .093 to .377 (Table 3). Evaluation-chosen patches scored 1.9 points above random patches ([−1.6, 5.3]). Exp5 repaired queries first and patched only graph-side failures, gating each patch on reserved curation CQs and never on held-out questions; evaluation-chosen patches again did not improve on random patches (−0.3; upper bound 3.1, margin 5).

A.6 Combined pipeline versus the loop

In Exp7 the combined pipeline and LOOP+vocab (Table 7) used 838k and 878k tokens in Qwen. Without a per-question triple cap the combined pipeline led in all three models, and at equal fact counts the lead shrank to 0.3 to 1.1 points (Table 16). In Qwen, capped recall favoured the loop (Table 3). On T-REx with Gemma, neither registered contrast was established (Table 11).

A.7 Further checks

Of the 2,196 gold evidence facts in the Exp2 graphs, 860 were reproduced exactly, 74 had a wrong value, 96 were split across entities and 1,166 were missing. Omissions made up 87.3% of the 1,336 inexact facts. SHACL shapes [31] flagged 0.15% of the inexact facts ([0, 0.37]) and 25.1% of correct facts.

Exp3: The fixed relation list scored higher than prompt optimisation with MIPROv2 or GEPA on 400 test questions, with and without matching the number of triples (Table 17, arm means).

Exp1: Iterative CQ refinement raised the gold ASK-query pass rate over no refinement on building-ontology questions (+21.4 points, [16.2, 27.1], 35 test questions; positive by construction).

TABLE 9
Exp8, other arms against the reference of 40 CQs at 1×, in points. The arms with 160 CQs have shard-cluster BCa 95% intervals; the 40-CQ arm at 4× is an arm-mean difference with no registered interval. Tokens are the mean ratio of 160 CQs at 4× to the reference.

	Qwen, T-REx	Gemma, T-REx	Gemma, 2Wiki	Qwen, 2Wiki
KG-only QA, 160 CQs, 4×	+4.7 [1.6, 8.0]	+5.0 [1.7, 8.2]	+10.6 [2.5, 17.2]	+1.6 [−4.7, 5.7]
KG-only QA, 40 CQs, 4× (mean)	+1.5	−0.5	+1.0	0.0
Evidence recall, 160 CQs, 4×	+1.9 [0.8, 3.9]	+0.6 [−0.2, 2.0]	+9.5 [5.9, 15.6]	+12.0 [8.8, 15.6]
Hybrid QA, 160 CQs, 4×	+0.9 [−2.8, 4.2]	+0.3 [−2.2, 2.6]	−0.9 [−4.7, 3.4]	−9.1 [−13.8, −5.5]
Precision, 160 CQs, 4×	−12.4 [−18.0, −7.8]	−1.2 [−4.8, 2.4]	+3.3 [−0.6, 7.1]	−12.2 [−17.0, −6.4]
Precision, 160 CQs, 1×	0.0 [−5.2, 5.0]	−0.5 [−5.2, 3.8]	+4.9 [0.8, 9.4]	−4.0 [−9.0, 0.7]
Tokens, 160 CQs 4× / reference	2.8	1.9	3.8	4.0

TABLE 10
Exp8: 160 CQs at the 1× token budget against the reference of 40 CQs at 1×. Differences in points with shard-cluster BCa 95% intervals; n is the number of shards in the contrast. Post hoc 90% intervals for hybrid QA, left to right: [−1.8, 2.3], [−1.7, 3.1], [−3.4, 4.7] and [−8.1, 0.3].

	Qwen, T-REx (n=33)	Gemma, T-REx (n=26)	Gemma, 2Wiki (n=20)	Qwen, 2Wiki (n=24)
Hybrid QA	+0.3 [−2.2, 2.6]	+0.6 [−2.2, 3.6]	+0.6 [−3.7, 5.6]	−3.6 [−8.9, 1.0]
KG-only QA	+3.5 [0.7, 6.8]	+5.6 [2.0, 9.8]	+10.9 [4.1, 15.9]	+6.2 [0.3, 10.4]
Evidence recall (uncapped)	+0.3 [−1.1, 2.1]	+1.0 [0.1, 2.3]	+9.2 [5.5, 15.4]	+9.9 [6.9, 13.2]

TABLE 11
Original Qwen runs and the Gemma replication, in points with 95% intervals (n in shards or units). Labels are the plans' decision labels; for Exp5 the margin is 5 points. A dash means no value or no run. Nemotron is a third model whose grounded QA stayed at floor (at most .047 in every arm, post hoc).

Experiment and contrast	Qwen	Gemma	Nemotron
Exp1, iteration over none (by construction, plan)	+21.4 [16.2, 27.1]	+42.1 [33.6, 50.7]	invalid or at floor
Exp4, GR over L0	+9.7 [4.1, 15.0]	+13.1 [7.8, 18.4]	invalid or at floor
Exp5, evaluation-guided over random repair	−0.3, upper bound 3.1 (one-sided), not improved	+1.5 [−3.1, 6.5], not improved	−0.7 [−1.9, 0.0]
Exp7, recall, combined over LOOP+vocab	+5.6 [4.2, 7.0]	+9.2 [7.6, 11.4] (n=60)	+13.1 [8.9, 16.8] (n=41)
Exp7, recall, combined over L0+vocab	+14.0 [12.1, 15.9]	+10.2 [2.1, 16.1]	+16.7 [11.8, 20.9]
Exp7 design on T-REx, combined − LOOP+vocab	—	+1.6 [−0.9, 4.4], not established (n=27)	+1.2 [−0.6, 4.1], not established (n=12)
Exp7 design on T-REx, combined − L0+vocab	—	+2.7 [−0.4, 5.8], not established	—

TABLE 12
Checks added after the main study, Qwen. Hybrid QA with up to 30 graph facts against vector-only QA (.502 [.460, .544]) and fact precision judged by Gemma, both on the 2Wiki Exp7 graphs (points, 95% intervals, 30 shards, 60 units); grounded QA, class-aligned recall and tokens per unit (thousands) for the Exp7 arms on T-REx (33 shards, one run seed). Precision was not measured for L0.

	Hybrid QA	Hybrid − vector	Precision	QA, T-REx	Recall, T-REx	Tokens (k)
L0	.600	+9.8 [4.3, 14.6]	—	—	—	—
L0+vocab	.692	+19.0 [14.7, 22.5]	.874 [.856, .890]	.290	.173	22.7
LOOP+vocab	.671	+16.9 [12.2, 21.0]	.802 [.785, .819]	.346	.189	310
Combined	.603	+10.1 [5.3, 14.2]	.716 [.686, .741]	.371	.228	298
U-ext+vocab	.585	+8.3 [2.2, 13.9]	.925 [.913, .938]	.421	.184	19.1

A.8 Exp8 arms and measures

The reference arm uses 40 CQs at the 1× token budget. For each shard the budget comes from a reference run, and an arm stops after using 95% of it. Other arms are named by CQ count and budget: 160 CQs at 1×, 160 CQs at 4× and 40 CQs at 4×. The source check is on in the 40-CQ and 160-CQ arms. Each generated CQ is embedded with Qwen3-Embedding-0.6B and compared with the top 5 retrieved chunks, and a CQ that fails is regenerated up to 3 times.

In Exp8, KG-only QA gives the answerer graph facts only. Measure F2 takes the vector answer unless it abstains, and then the KG-only answer.

A.9 Registration records and run details

Plans and labels: Each experiment has a plan, fixed before the main run, that states hypotheses and decision rules, and a deviation log that we kept during the run. The list comparisons in Exp7 are descriptive intervals, and G−L0 in Exp4 and Exp5 (Table 1) is a post hoc secondary contrast. Seeded extraction (U-ext+vocab) is an arm added after an internal review and run under its own addendum. The comparison of iText2KG with the Exp7 arms was registered, and its outputs are descriptive; the comparisons with EDC and AutoSchemaKG were registered as descriptive, except EDC with the list (registered, Holm-corrected). Decisions use the labels of each plan: *improved*, *not improved* and *not established*. An interval that includes zero is reported as *did*

TABLE 13
The original list (relation names taken from the 2WikiMultihopQA training split) the induced list (proposed by an LLM from 300 corpus chunks), and a list induced the same way from 300 chunks outside the evaluation shards.

Original	award received, cause of death, child, composer, country, country of citizenship, country of origin, date of birth, date of death, director, educated at, father, founded by, inception, mother, occupation, performer, place of birth, place of burial, place of death, publication date, spouse
Induced	award, based on, birth date, birth place, cast member, child, country, death date, death place, director, employer, genre, located in, member of, nationality, occupation, parent, position held, release year, sibling, spouse, title
Induced outside shards	based on, birth date, birth place, cast member, child, death date, death place, director, employer, genre, location, member of, nationality, occupation, parent, producer, publication date, release year, sibling, spouse, title, writer

TABLE 14
Post hoc list effect in points (95% BCa intervals, 30 shards, 60 units), Exp7. Class-free recall ignores relation classes when matching triples to gold; answer correctness drops the grounding requirement.

	Recall, class aligned	Class-free recall	Answer correct
L0	+11.4 [9.3, 14.5]	+9.7 [7.6, 12.9]	+13.8 [10.5, 17.2]
LOOP	+7.7 [5.4, 9.7]	+5.0 [2.9, 6.9]	+7.9 [4.8, 10.7]
Combined	+10.3 [8.0, 12.6]	+6.8 [5.1, 8.6]	+7.0 [4.0, 9.7]
U-ext	+2.1 [1.0, 3.5]	+0.6 [−0.7, 2.3]	+3.0 [0.0, 6.6]

TABLE 15
Guided single pass minus a selection of the extraction graph at the guided pass's fact count (Qwen on 2Wiki, 60 shard × run seed units; grounded QA, differences in points with 95% intervals). Status gives the outcome of the registered test of whether the selection scores above the guided pass. The selectors read no questions or gold relations; one arm reads the unit's CQs.

Selection	Relation list	Guided − selection	Status
Entity centrality	none	+14.4 [9.5, 19.5]	Reg.: guided higher
Relation frequency	none	−8.0 [−13.3, −2.7]	Reg.: selection higher
Entity centrality	seeded	+25.8 [20.5, 30.5]	Reg.: guided higher
Relation frequency	seeded	+16.6 [11.4, 22.1]	Reg.: guided higher
Chain-preserving	seeded	+25.0 [19.8, 30.1]	Reg.: guided higher
Model-scored	seeded	+14.3 [8.9, 19.3]	Reg.: guided higher
CQ-selected	seeded	+33.3 [28.3, 37.7]	Reg.: guided higher
Chunk-balanced	seeded	+37.2 [31.6, 41.4]	Reg.: guided higher

TABLE 16
Exp7: recall lead of the combined pipeline over LOOP+vocab, in points with 95% intervals; uncapped means without a per-question triple cap. Intervals resample shard × run seed units by shard.

Model	Units	Uncapped recall lead	At equal fact counts
Qwen	60	+5.6 [4.2, 7.0]	+1.1 [0.5, 1.9]
Gemma	60	+9.2 [7.6, 11.4]	+0.3 [−0.6, 1.1]
Nemotron	41	+13.1 [8.9, 16.8]	+0.3 [−0.5, 1.3]

TABLE 17
Exp3 evidence recall on 400 test questions. The last column equalises the number of triples across arms.

	Recall	Recall, equal triples
Relation list	.425	.411
MIPROv2	.363	.255
GEPA	.356	.337
Default prompt	—	.181

not separate. The plan for the chunk-retrieval comparison registered answer correctness as the measure.

Checks added after the main study: Eight checks added after the main study each had a plan fixed before its run: the synonym list, fact precision, vector versus hybrid QA, the Exp7 arms on T-REx, the subsets at equal fact count, the induced list, the list induced outside the evaluation shards and the share of the list given. A further check compares chunk retrieval with the graph. The synonym-list check replaces the 22 names by synonym phrases; its hypotheses are registered and its vocabulary-adoption measure is post hoc. Its comparison arms are the stored Exp7 runs, and a rerun of four arms on the same instance reproduced the stored values. Hypotheses and analysis were fixed before each check. Fact precision has a registered non-inferiority contrast (margin -5 points). On T-REx, the Exp7 arms have a registered equivalence test (margin ± 3 points); seeded extraction led the combined pipeline there, so the test did not establish equivalence. In Exp8, the hybrid-QA contrasts are registered and only the analysis of why hybrid QA fell is post hoc; KG-only QA and evidence recall are secondary measures.

Serving instances: The 60-fact hybrid run, the outside-shard list run and the EDC run with the list used another serving instance than the stored arms they are compared with; the instance changed during the hybrid and outside-shard runs and after the first 12 EDC units. The same arm differed between serving instances (no list .433 stored, .417 rerun), so contrasts are taken within one instance where possible. The retrieval comparison reran seeded extraction on the retrieval arms' own serving engine (.669 against .678 in the stored run). The selector arms of the equal-fact-count check ran on a different serving engine from the stored guided-pass arm. The reruns of the dose check (Table 5) all come from one instance; the stored Exp7 values come from another.

List induction: For the induced list an LLM proposed relation names from 300 chunks of the corpus being built and the proposals were merged into 22 names. No question and no gold relation was used. On T-REx a list of 22 names was induced the same way from 300 of the 386 evaluation-corpus passages, again with no questions and no gold relations. The induction therefore read passages of the corpus that is later queried (a transductive setting), and 4 of its 22 names match the 2Wiki list exactly. The list induced outside the evaluation shards was built with the same procedure from 96 passages outside the evaluation shards, so no evaluation passage and no question was read; 15 of its 22 names match the first T-REx list and 6 match the 2Wiki list. It gave +7.4 points of grounded QA over no list in the single pass (Section 4).

Statistics: Intervals resample 30 shards on 2Wiki (two run seeds averaged within a shard, 60 units) and 33 shards on T-REx per run seed. Intervals are two-sided 95% unless marked one-sided or post hoc 90%. P-values from 10,000 resamples have a floor of 1/10,001. Post hoc analyses reuse the stored predictions and make no new model calls.

Status of contrasts reported in Sections 4 to 6: Registered, improved, with Holm-adjusted p: synonym list in L0, answer correctness (0.0002); induced list in L0, answer correctness (0.001); outside-shard list on T-REx, grounded QA (0.0001); outside-shard list on 2Wiki, grounded QA (0.0002); the T-REx single pass against an equal-size random

subset of seeded extraction (0.001); guided pass against the chain-preserving selector and the model selector (both 0.0002); against the selection by the unit's CQs (0.0004) and the chunk-balanced selection (0.0005); seeded extraction against EDC with the list (0.0003); the T-REx single pass with the induced list over no list, grounded QA and answer correctness, in both run seeds (Table 4). Registered, improved at the boundary: extraction from every chunk with the outside-shard 2Wiki list, grounded QA (+3.1 [0.0, 6.5]). Registered, not established: extraction from every chunk with the outside-shard 2Wiki list, answer correctness (+1.7 [−1.0, 4.2]); EDC with against without the list (+0.4 [−2.7, 2.9]); on T-REx without the list, the guided pass against a random subset (6.6 [−5.8, 12.2]). In the dose check, 11 of 22 names against no list is registered, improved (+10.5 [5.5, 14.5]); the dip at 6 names is a post hoc observation. In the equal-fact-count check without the list, the contrast of the relation-frequency selection above the guided pass (+8.0 [2.7, 13.3]) is registered. The gain of U-ext over L0 without the list (+16.5 [10.6, 21.7]) is post hoc. Seeded extraction against the combined pipeline on 2Wiki is an added arm with a descriptive interval; the answer-correctness contrast (+2.3 [−0.2, 4.4]) and its 90% interval [−3.0, 2.6] are post hoc. The comparisons of public extractors with Exp7 arms, the rerun of the original list beside the induced list, the reach rates and the context-free answer model are descriptive; the relation-class requirement and the relaxed entity linking are post hoc reanalyses of the stored predictions and graphs. Chunk-retrieval comparisons (graph against chunk retrieval and list-hinted retrieval) are registered, improved; the raise of the fact cap from 30 to 60 in hybrid QA (U-ext+vocab, Combined) is registered, improved in answer correctness.



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